



Understanding the Relationship Between Rodent Violations and Extreme Temperatures

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Research Question:

What is the relationship between the number of rodent-related health code violations and the frequency of heat advisory warnings (extreme summer temperatures) in Boston from 2007-2025?

Background:

- Increase of extreme temperatures during summer due to climate change.
- Rats actively avoid extreme heat, seeking shelter in human spaces.
- This causes restaurants to be at risk, as they have an abundance of food, water, and shelter.

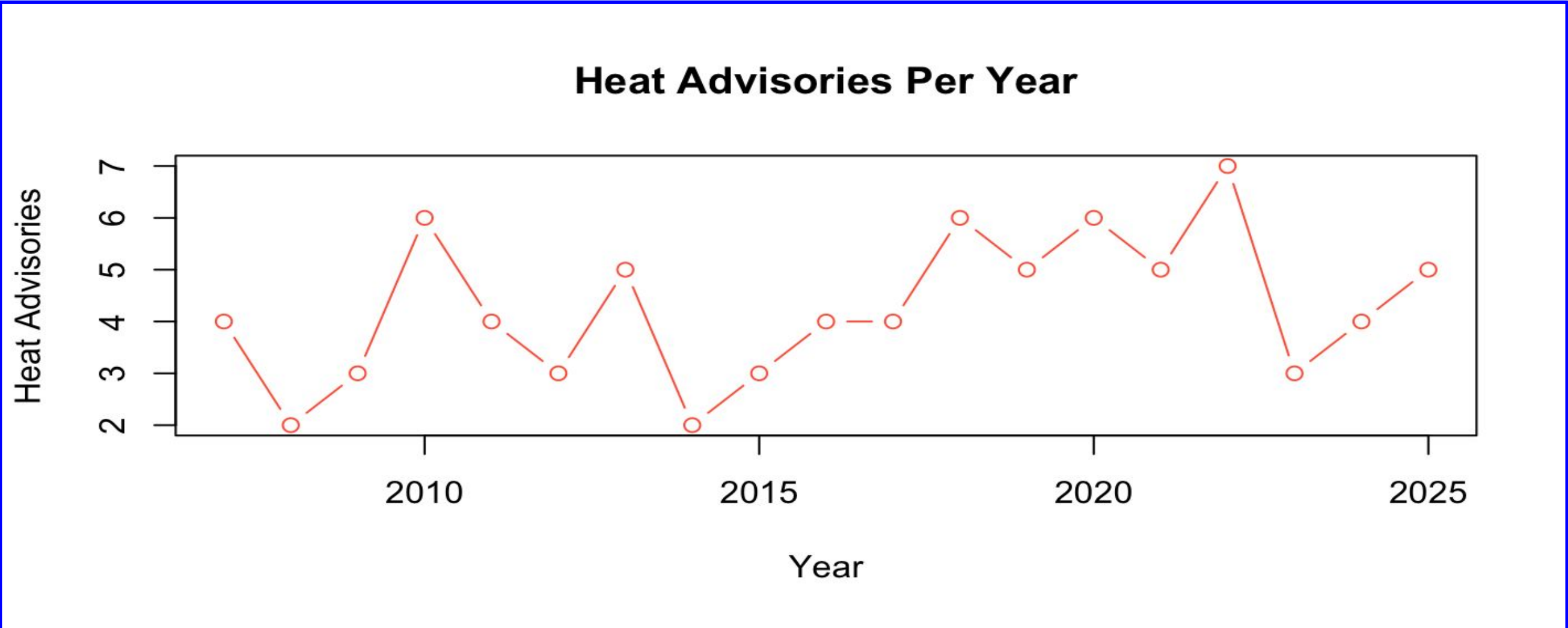


Figure 1: Scatterplot of heat advisories over time

Data & Methods:

Primary Data Source:

- Boston Food Inspections; **Attributes:** violation time, comments
- NOAA Climate Data for Boston ; **Attributes:** max temp

Data Manipulation:

- Created rodent_rate (total violations in a day / total inspections in a day)
 - Characterized by “rodent” in comments.
- Created heat_wave_start (if it was the beginning of a heat advisory – 3 days of 85° or more).
- If the number of violations were outliers, that observation was removed.
- Merged these two datasets using the “date” column as a primary key.
- If a food inspection row did not contain the violation time, it was dropped.
- For the logistic regression, any_rodent was created to signify if a rodent violation was found on a day.
- For the beta regression, only dates with rodent rates that were greater than 1 were included.
- 80% train, 20% test split

Visualizations:

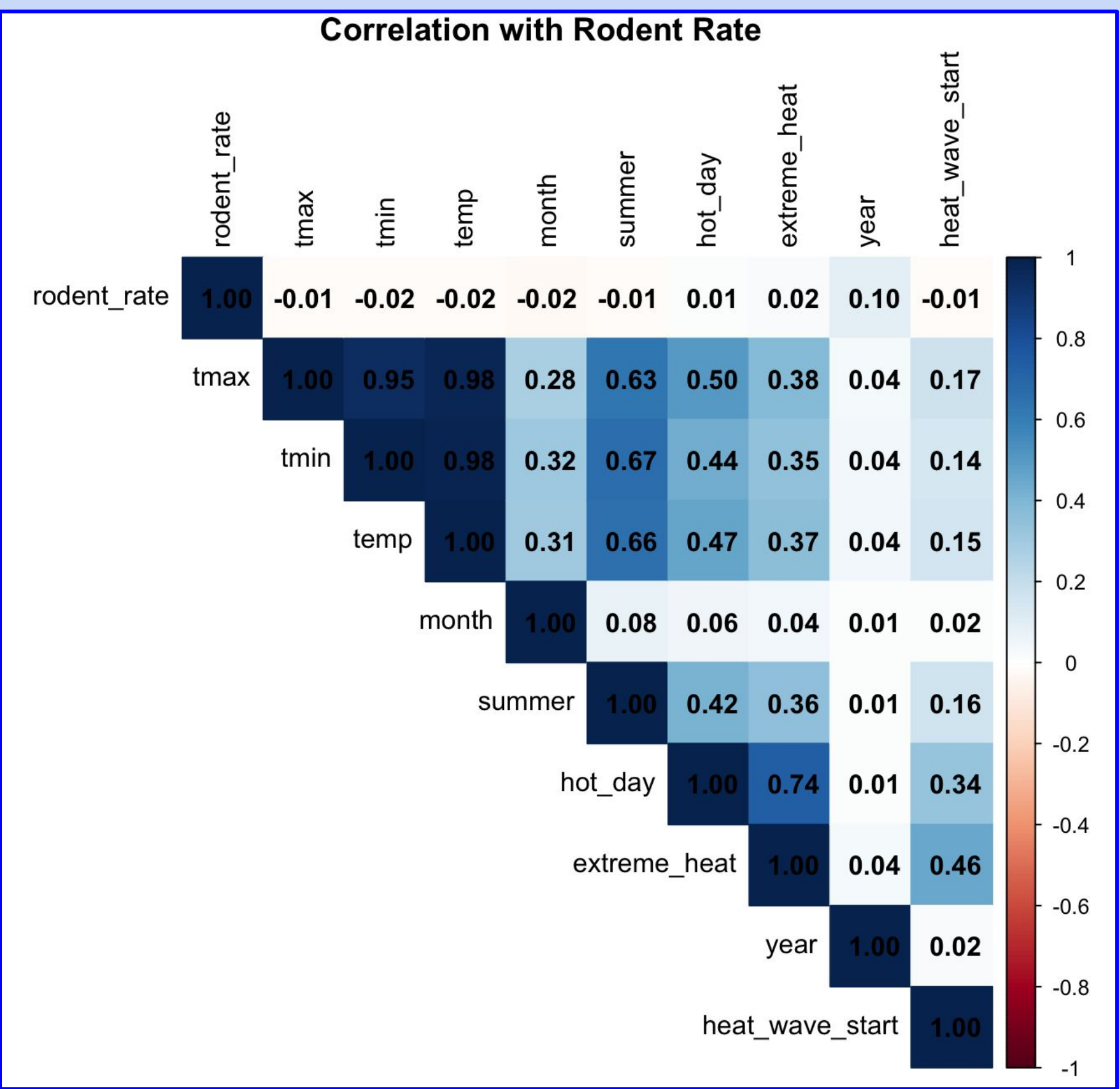


Figure 2: Correlation plot of all potential variables.

- Relationship between our response (rodent rate) and our explanatory variables are very weak.
- The strongest correlation is between rodent rate and year, or a measure of time.
- The heat related variables do not show strong correlation at all.

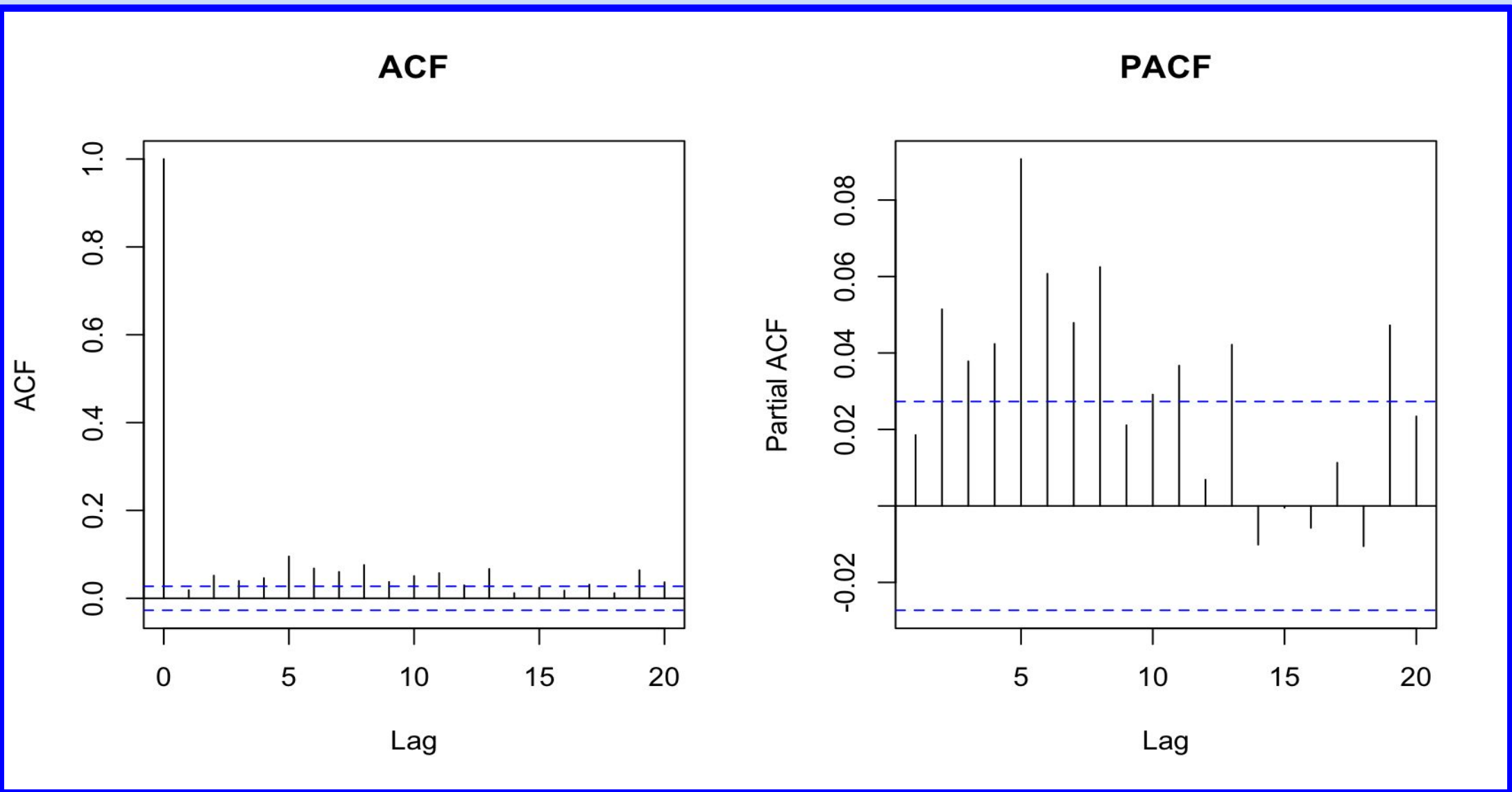


Figure 3: ACF and PACF plots

Modeling:

- Plotted ACF & PACF plots to determine potential lags and stationarity in data
- The ACF plot shows that there is not much signal between the current value and past values.
- The PACF plot shows a strong signal at lag 5, hinting at potential correlation between the current value and the value 5 steps ago.
- However, when comparing the model with a lag 5 steps ago vs the model with no lag, the AIC values differed by less than 0.01.

Model 1: OLS Regression with no autocorrelation

Model 2:

Part 1: Logistic Regression – Any rodent violations on a specific day?

Part 2: Beta Regression – Given that violations actually occurred, what rodent rate can be predicted?

Results:

	Model 2, Pt. 1: Logistic Regression	Model 2, Pt. 2: Beta Regression
Date	1.456e-04 p-value: < 2e-16 ***	0.0082734 p-value: 0.378
Max Temperature	-9.999e-03 p-value: < 0.00275 **	-0.0011941 p-value: 0.201
Heat Wave Start?	1.179e-02 p-value: 0.97038	-0.0559124 p-value: 0.546
Combined RMSE	0.0848	

	Logistic Regression	Beta Regression
Regressions without heat variables	p-value: 0.009937 ***	p-value: 0.3154

Logistic Regression (Part 1): p = 0.009937 ***

- Highly significant → Heat variables are helpful for predicting whether you get a zero.

Beta Regression (Part 2): p = 0.3154

- Not significant → Heat variables don't matter for predicting the magnitude among non-zeros

** On average, predictions are off by ~0.085 in rodent rate

Conclusion:

- Heat advisories **do not appear** to associate with the increase rodent violation rates in Boston from 2007 to 2025
- Notable: The **negative tmax in Part 1**; This suggests that inspections may be suppressed on hot days.
- Appears that for reaching the threshold to have a rodent violation, heat does play a role.
 - However, when determining the magnitude of the rate, heat does not appear to be correlated.

Limitations

Exclusion of dates – outliers

- Data quality issues?
- Detection != violation occurrence time
- Space correlation?

Discussion & Future Work

- Contrasts the existing research done on rodent sightings and extreme temperature.
 - Heat advisories may not be the only thing to be focusing on (humidity, rain, etc)